



Instituto Politécnico
de Castelo Branco



Instituto Politécnico de Castelo Branco
Escola Superior de Tecnologia

Cycle in **Degree** Course
2nd Semester of 3rd Year
Course Unit: **Cloud Computing**
Lecturer: Osvaldo Santos

Practical work on Azure App Service

Objectives of this sheet

Initial contact with the Azure App Service

Bibliography and useful links

Azure documentation on App Service

<https://learn.microsoft.com/en-us/azure/app-service/>

Azure App Service CI/CD

[Configure continuous deployment - Azure App Service | Microsoft Learn](#)

How to configure a Node application for Azure App Service

<https://learn.microsoft.com/en-us/azure/app-service/configure-language-nodejs?pivots=platform-linux>

How to create a personal access token on GitHub

<https://docs.github.com/en/authentication/keeping-your-account-and-data-secure/creating-a-personal-access-token>

Practical work on Azure App Service

In this job you will use your ipcbcampus student account to access Azure

At the end of the job, you must delete all Azure resources so that you can preserve your balance

In this work, you will use source code from a supposed bank website (Insecure Bank), code that is used in computer security classes at ESTCB, because it purposely has a number of application vulnerabilities. In the context of this work, the code is not really functional, it only serves to show the initial login page. So there's no point in trying to log in.

Part I - A web application, in PHP

1) PHP source code preparation

- a) In this work we're going to use GitHub as a source code repository. If you don't already have a GitHub account, create a new one.
- b) On your computer (or virtual machine), create a folder called **insecure_bank_php** and enter this folder. Download and unzip the project's source code from moodle. The file in moodle should be called **insecure_bank_php.zip** or similar.
- c) Next, on the GitHub website, create a new **private** repository in your GitHub account, called **insecure_bank_php**, and upload the files with the source code (which you have just downloaded) to GitHub.

Tip:

It can be complicated to upload more than 100 files to GitHub using the web interface of the GitHub website. An alternative is to use GitHub Desktop or the git CLI (command line).



Instituto Politécnico
de Castelo Branco

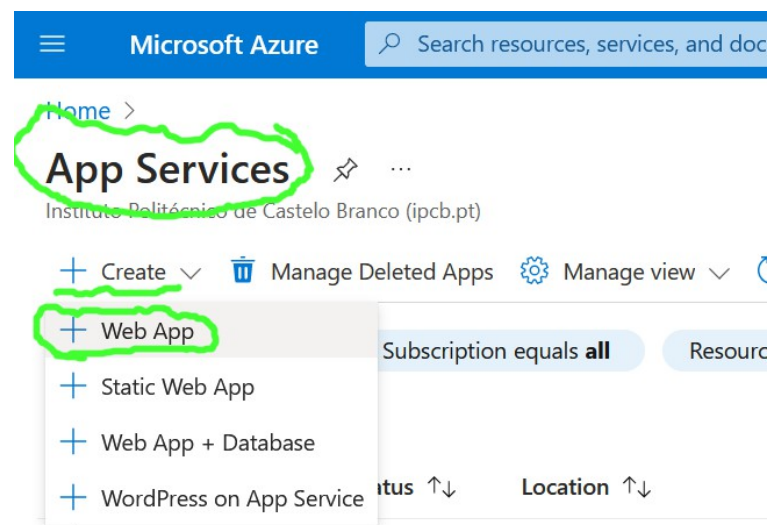


Instituto Politécnico de Castelo Branco
Escola Superior de Tecnologia

School of Technology
Computer Engineering course
Cycle in **Degree** Course
2nd Semester of 3rd Year
Course Unit: **Cloud Computing**
Lecturer: Osvaldo Santos

Practical work on Azure App Service

2) Creation of the App Service application in PHP, from GitHub



a) Create a new application in the App service (Web App), with the following characteristics:

i) Basics

- Resource Group: class
- Application name: app-"first_name"-"last name", for example "app-joao-ronaldo". If the name is already in use, use an alternative name
- Publish: Code
- Runtime Stack: PHP 8.2 (or higher) on Linux
- Region: Uk South
- Pricing Plan: Basic B1

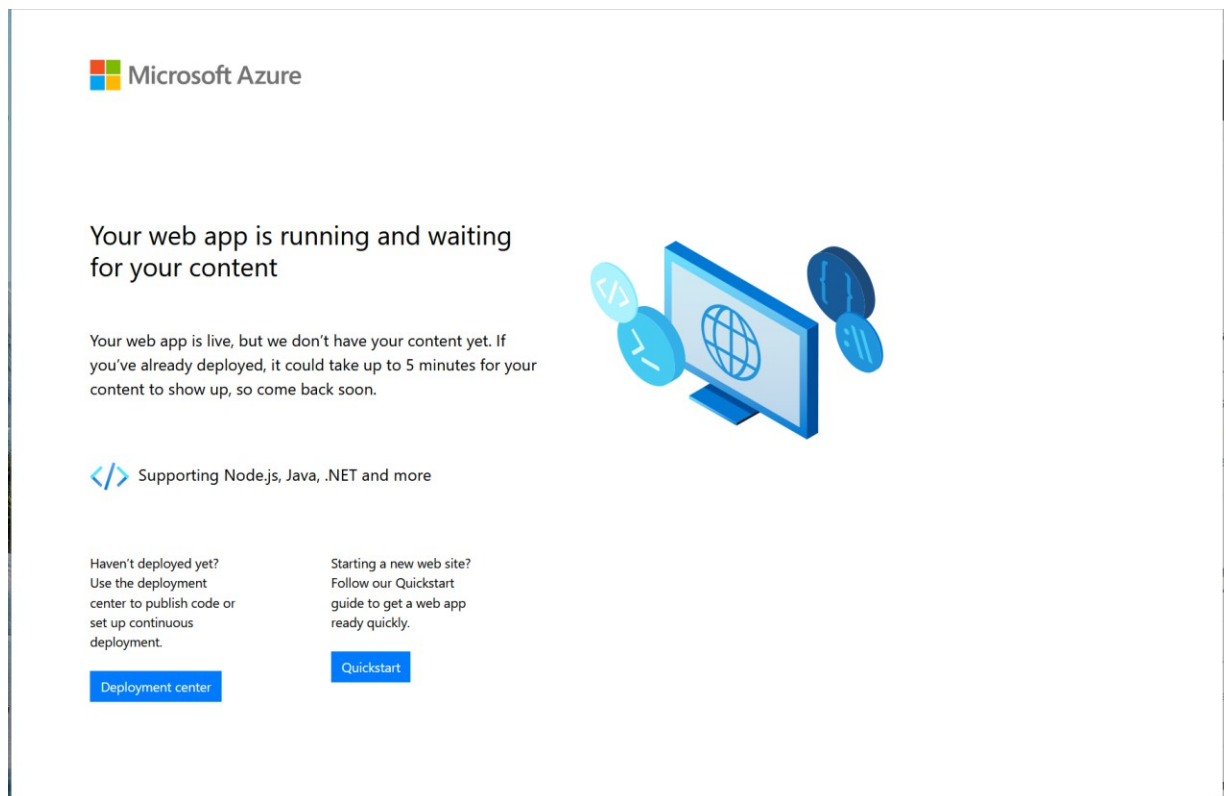
ii) Deployment

- GitHub Actions settings, Continuous deployment: Disable
- GitHub account: select your GitHub account

iii) You don't need to change anything in the other sections.

Practical work on Azure App Service

- iv) You can create the application (Create)
- b) Wait until the application is created and go to the resource.
- c) What is the URL of the application? If you open the URL in a browser, what do you see? Wait a few minutes for the application to come online; after a few minutes, you should see something like this:



- d) Now, in Azure, in the App Service, go to the "Deployment Center" section and deploy the PHP project from the repository of your GitHub account, branch main, option add a workflow

Note: if you have problems integrating with GitHub, please refer to Annex I at the end of the statement.



Instituto Politécnico
de Castelo Branco

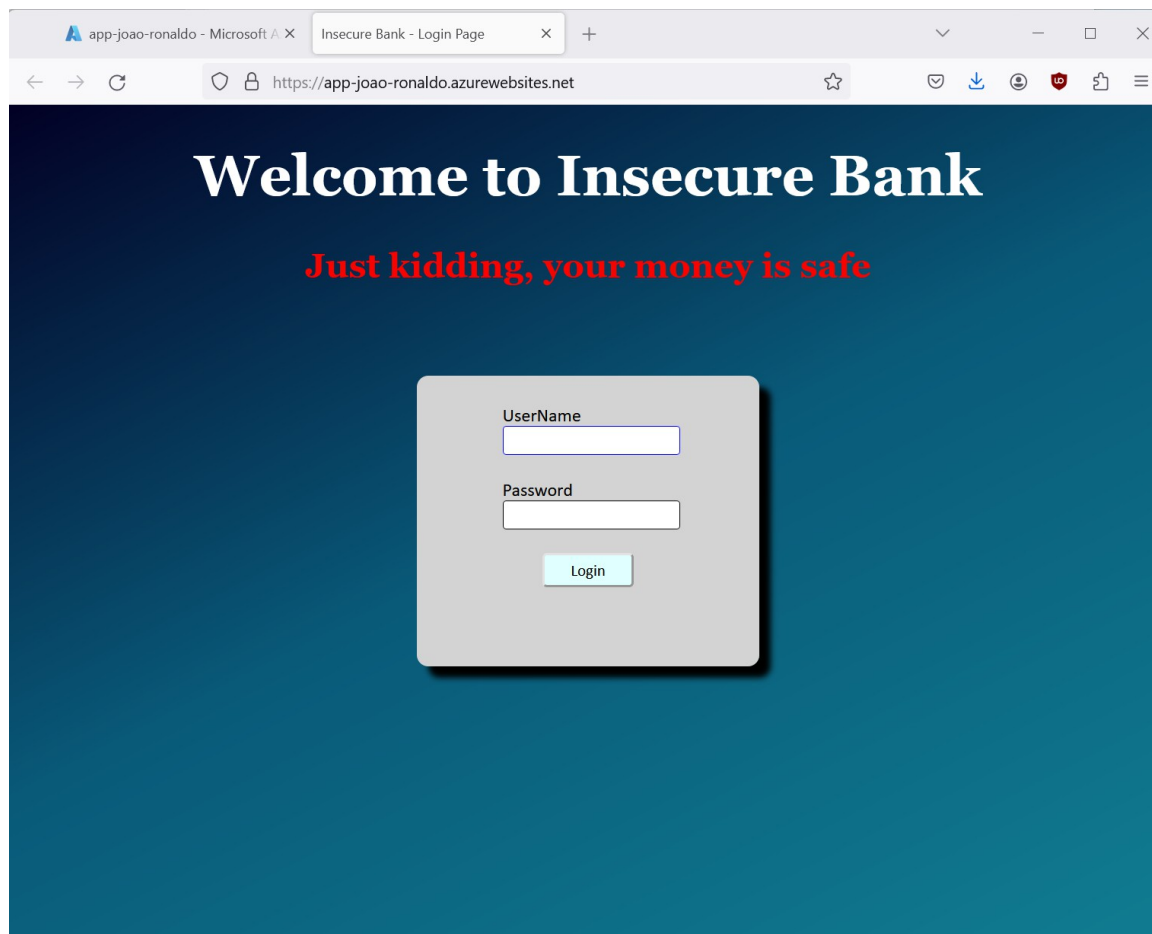


Instituto Politécnico de Castelo Branco
Escola Superior de Tecnologia

School of Technology
Computer Engineering course
Cycle in **Degree** Course
2nd Semester of 3rd Year
Course Unit: **Cloud Computing**
Lecturer: Osvaldo Santos

Practical work on Azure App Service

- e) Wait until the deployment is complete (it may take a few minutes) and refresh the project page. You should see something like this:



- f) Notice one detail: is the website insecure (HTTP) or secure (HTTPS)? If you look at the security of the website (browser padlock symbol) what is reported?
- g) Now delete this application in the App Service. The best way is to delete the resource group.



Instituto Politécnico
de Castelo Branco



Instituto Politécnico de Castelo Branco
Escola Superior de Tecnologia

School of Technology
Computer Engineering course
Cycle in **Degree** Course
2nd Semester of 3rd Year
Course Unit: **Cloud Computing**
Lecturer: Osvaldo Santos

Practical work on Azure App Service

Part II - Another web application, now in NodeJS

3) Source code preparation in Node.JS - using the git CLI

- a) On GitHub, create a new private repository called **insecure_bank_node**, and leave it empty.
- b) Install the Git CLI software on your computer (or virtual machine)
- c) On your computer (or virtual machine), create a folder called **insecure_bank_node** and enter this folder. Download the project's Node.JS source code from moodle. The file in moodle should be called **insecure_bank_node.zip** or similar.
- d) Now, from this folder, you will upload the files to your repository on GitHub. Follow these steps:
 - i) Check that it is in the **insecure_bank_node** folder
 - ii) Initialize the repository / folder

```
git init -b main
```

- iii) Now associate your email with this project (use the same one you used on GitHub)

```
git config --global user.email "you@example.com"
```

- iv) Now associate this project with your name

```
git config --global user.name "Your Name"
```

- v) Now add all the files, and then make the commit

```
git add .
```

- vi) Now commit the project in its current state

```
git commit -m "First commit"
```

- vii) Now link this local repository to your repository on your GitHub account. The command should all be on the same line and the last parameter, the URL, should be the URL of your GitHub repository



Instituto Politécnico
de Castelo Branco



Instituto Politécnico de Castelo Branco
Escola Superior de Tecnologia

School of Technology
Computer Engineering course
Cycle in **Degree** Course
2nd Semester of 3rd Year
Course Unit: **Cloud Computing**
Lecturer: Osvaldo Santos

Practical work on Azure App Service

```
git remote add origin https://github.com/"replace with  
your git username"/insecure_bank_node.git
```

viii) Finally, upload the files to the GitHub repository. If you are asked for a username and password, log in via your browser.

```
git push origin main
```

ix) Check GitHub to see if the files have been well received.

x) Do not close the local folder where you are working.



Instituto Politécnico
de Castelo Branco



Instituto Politécnico de Castelo Branco
Escola Superior de Tecnologia

School of Technology
Computer Engineering course
Cycle in **Degree** Course
2nd Semester of 3rd Year
Course Unit: **Cloud Computing**
Lecturer: Osvaldo Santos

Practical work on Azure App Service

4) Creation of the application in Node.js, with CI/CD - Continuous Integration / Continuous Delivery

a) Create a new application in the App Service (Web App), with the following characteristics:

i) Basics

- Resource Group: class
- Application name: app-"first_name"-"last name", for example "app-joao- ronaldo". If the name is already in use, use an alternative name
- Publish: Code
- Runtime Stack: Node 20 LTS on Linux
- Region: Uk South
- Pricing plan: Basic B1

ii) Deployment

- GitHub Actions settings, Continuous deployment: Enable
- GitHub account: your personal GitHub account (you may need to log in)
- Repository: o repository that just created created in your account GitHub (insecure_bank_node)
- Branch: main

iii) You don't need to change anything in the other sections.

iv) You can create the application (Create)



Instituto Politécnico
de Castelo Branco

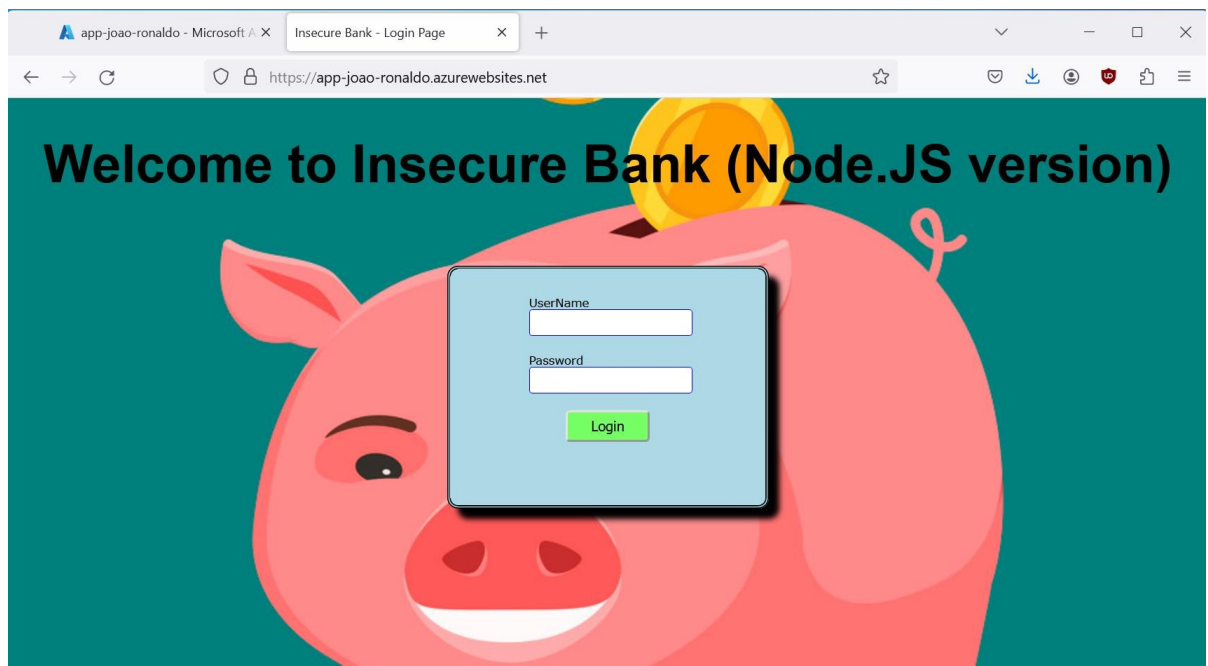


Instituto Politécnico de Castelo Branco
Escola Superior de Tecnologia

School of Technology
Computer Engineering course
Cycle in **Degree** Course
2nd Semester of 3rd Year
Course Unit: **Cloud Computing**
Lecturer: Osvaldo Santos

Practical work on Azure App Service

- b) Wait until the deployment is complete and refresh the project's web page. You should see something like this:





Instituto Politécnico
de Castelo Branco



Instituto Politécnico de Castelo Branco
Escola Superior de Tecnologia

School of Technology
Computer Engineering course
Cycle in **Degree** Course
2nd Semester of 3rd Year
Course Unit: **Cloud Computing**
Lecturer: Osvaldo Santos

Practical work on Azure App Service

5) CI/CD tests

- a) First of all, you need to update the local version of the project. In the project's local folder, run the following commands. Note: in the second command, you must use the identifier of your repository.

```
git config pull.rebase true  
git pull https://github.com/"replace"/insecure_bank_node.git main
```

- b) Now, edit the public/index.html file, and change the message in the html code to "Welcome to Bank XPTO".
- c) On the GitHub site, in the insecure_bank_node repository, open the "Actions" section and note what is there. Keep this page visible while you carry out the next operations.
- d) Now commit the changes and push them to the local folder on your computer.

```
git add .  
git commit -m "Modified index.html"  
git push origin main
```

- e) Now take a look at the actions section of GitHub. What's happening?
- f) Wait a few minutes and refresh your web application in the browser. What happened?
- g) What is the advantage of this type of automation?
- h) Don't delete this project, you'll need it next.



Instituto Politécnico
de Castelo Branco



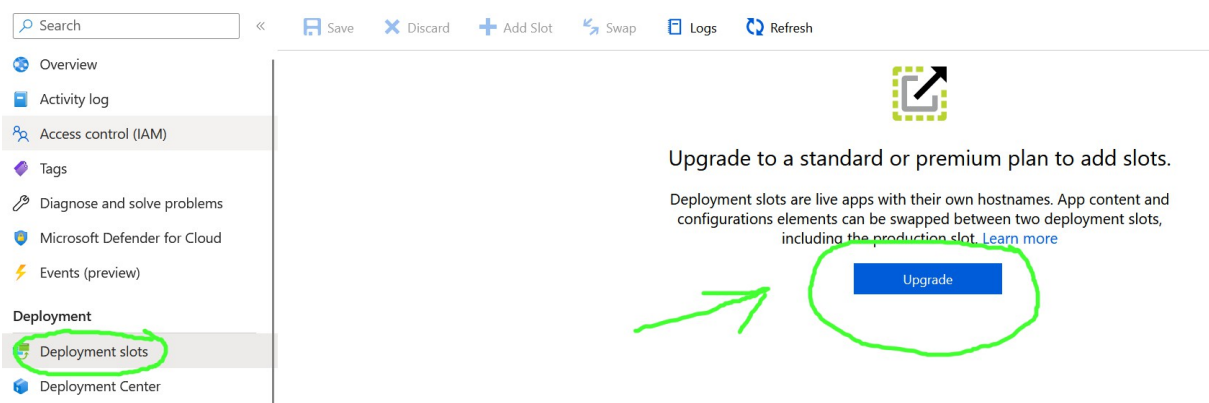
Instituto Politécnico de Castelo Branco
Escola Superior de Tecnologia

School of Technology
Computer Engineering course
Cycle in **Degree** Course
2nd Semester of 3rd Year
Course Unit: **Cloud Computing**
Lecturer: Osvaldo Santos

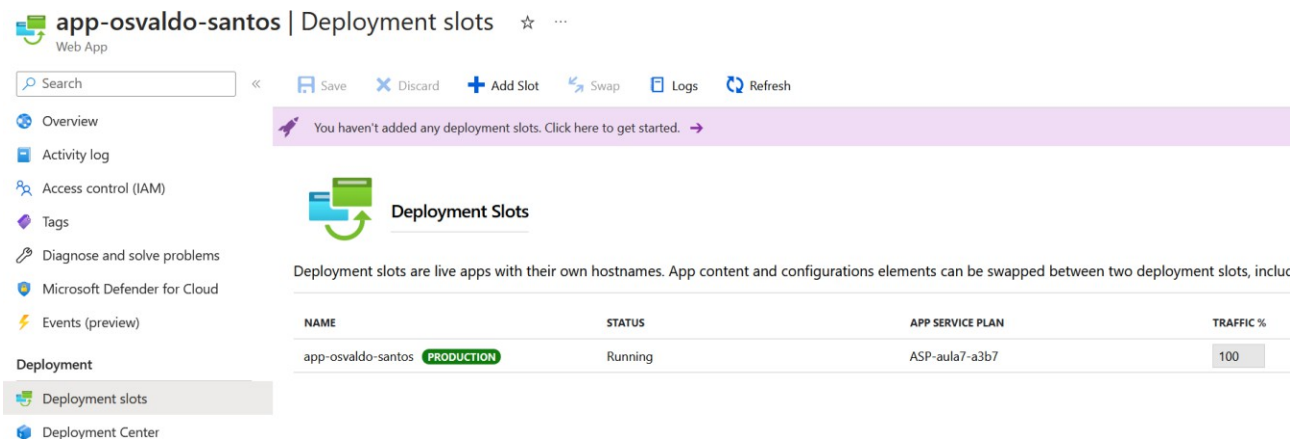
Practical work on Azure App Service

6) Deployment slots

- a) In Azure, in App Service, in the "Deployment slots" option, upgrade to **Production/ Premium V2 P1V2** to allow multiple deployment slots.



- b) Go back to the "deployment slots" option. The interface should now show something like this gender:





Instituto Politécnico
de Castelo Branco



Instituto Politécnico de Castelo Branco
Escola Superior de Tecnologia

School of Technology
Computer Engineering course
Cycle in **Degree** Course
2nd Semester of 3rd Year
Course Unit: **Cloud Computing**
Lecturer: Osvaldo Santos

Practical work on Azure App Service

- c) Add a new slot called "tests", without cloning the settings from the production slot. You should end up with a scenario similar to this:

The screenshot shows the Azure portal interface for the resource 'app-joao-ronaldo' under the 'Deployment slots' section. The left sidebar contains navigation links: Overview, Activity log, Access control (IAM), Tags, Diagnose and solve problems, Microsoft Defender for Cloud, Events (preview), and Deployment. The main content area displays the 'Deployment Slots' configuration. A table lists the following slots:

NAME	STATUS	APP SERVICE PLAN	TRAFFIC %
app-joao-ronaldo	PRODUCTION	ASP-aula-be26	100
app-joao-ronaldo-tests	Running	ASP-aula-be26	0

- d) Check the URL of the application in this test slot and try opening it in a browser. What comes up?
- e) Now, on GitHub, in the **insecure_bank_node** repository, create a branch called tests.
- f) Still on GitHub, in this testing branch, change the welcome message and even the background image of the website.
- g) In Azure, in the deployment slot tests, deploy the website to this new slot, from GitHub, branch tests. Wait a few minutes.
- h) Refresh the URL of the test slot application. What do you see now?
- i) Now swap the test slot with the production slot. Wait for the operation to complete.
- j) Open the URL of the production slot (the original URL of the bank's website) in a browser. What happened?
- k) What is this feature of having several deployment slots and allowing swapping between them useful for?

Practical work on Azure App Service

7) Conclusion

- a) What are the advantages and disadvantages of this application App Service compared to the alternative of creating a web server on a virtual machine?

Now delete all resources

Annex I - troubleshooting integration problems with GitHub

The implementation of CI/CD between the Azure App Service and GitHub requires access authorization from GitHub to the Azure application service. In principle, this authorization should be automatic, as soon as the GitHub account is selected in Azure.

If problems arise, start by logging out of both accounts, clearing all browser history, and trying again.

If the problems persist, on GitHub, check the user's Settings, the Applications section

Integrations

☰ Applications

🕒 Scheduled reminders

If the "Azure App Service" application is authorized.



Instituto Politécnico
de Castelo Branco



Instituto Politécnico de Castelo Branco
Escola Superior de Tecnologia

School of Technology
Computer Engineering course
Cycle in **Degree** Course
2nd Semester of 3rd Year
Course Unit: **Cloud Computing**
Lecturer: Osvaldo Santos

Practical work on Azure App Service

Applications

Installed GitHub Apps

Authorized GitHub Apps

Authorized OAuth Apps

You have granted **1 application** access to your account.

Sort ▾

Revoke all



Azure App Service Creates

Last used within the last week · Owned by [AzureAppService](#)

...

If the application is authorized and the problem still persists, try revoking this authorization and authenticating again in Azure.

In the repository, delete the .github/workflows folder